

# Incident Lifecycle Automation in ServiceNow

## Project Documentation Compilation

### 1. Salesforce — Model Performance Test

|                      |                                             |
|----------------------|---------------------------------------------|
| <b>Date</b>          | 08 September 2026                           |
| <b>Team ID</b>       |                                             |
| <b>Project Name</b>  | Incident Lifecycle Automation in ServiceNow |
| <b>Maximum Marks</b> | 10 Marks                                    |

#### Model Performance Testing

| S.No. | Parameter        | Values                                                                                                                                                                                                                                                                             |
|-------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | Model Summary    | Salesforce integration used to sync Account and Contact records into ServiceNow, giving Service Desk agents customer/account context on incidents linked to the affected Service. On sync, matching Account/Contact records are updated; unmatched records are flagged for review. |
| 2     | Accuracy         | Sync Match Accuracy – 98% (Account/Contact records correctly matched to existing ServiceNow CI/Contact records)                                                                                                                                                                    |
| 3     | Confidence Score | Reflects how confident the matching logic is when linking a Salesforce record to an existing ServiceNow record. Average confidence – 92%.                                                                                                                                          |

2. Artificial Intelligence — Model Performance Test

|               |                                             |
|---------------|---------------------------------------------|
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| Maximum Marks | 10 Marks                                    |

Model Performance Testing

| S.No. | Parameter                    | Values                                                                                                                                                                                                                                      | Screenshot |
|-------|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| 1     | Model Summary                | ServiceNow Predictive Intelligence model trained on historical incidents to auto-suggest Category, Subcategory, and relevant Knowledge Articles from the incident’s short description, reducing manual triage time for Service Desk agents. | N/A        |
| 2     | Accuracy                     | Training Accuracy – 91% / Validation Accuracy – 88% (suggested Category/Article matched agent’s final selection)                                                                                                                            | N/A        |
| 3     | Fine Tuning Result (if Done) | Validation Accuracy – 93% after retraining the solution on 3 months of additional resolved incidents                                                                                                                                        | N/A        |

3. Machine Learning — Model Performance Test

|               |                                             |
|---------------|---------------------------------------------|
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| Maximum Marks | 10 Marks                                    |

Model Performance Testing

| S.No. | Parameter      | Values                                                                                                                                                                                                                                                                                                               | Screenshot |
|-------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| 1     | Metrics        | Underlying classification solution for the Predictive Intelligence model (Category / Subcategory / Urgency auto-assignment).<br><b>Classification Model:</b><br>Confusion Matrix – 92/100 correctly classified on test set, Accuracy Score – 92%, Classification Report – Precision 0.91, Recall 0.90, F1-score 0.90 | N/A        |
| 2     | Tune the Model | Hyperparameter Tuning – adjusted cluster similarity threshold and minimum cluster size in the Solution Definition / Validation Method – 5-fold Cross-Validation on historical incident data                                                                                                                          | N/A        |

## 4. Power BI — Model Performance Test

|                      |                                             |
|----------------------|---------------------------------------------|
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| <b>Maximum Marks</b> | 10 Marks                                    |

### Model Performance Testing

| S.No. | Parameter                   | Screenshot / Values                                                                                                             |
|-------|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| 1     | Data Rendered               | Incident, Service, SLA, and related-record data pulled from ServiceNow via the built-in OData/Table API connector for reporting |
| 2     | Data Preprocessing          | Filtering and structuring incident records by Category, Status, Priority, and Assignment Group                                  |
| 3     | Utilization of Data Filters | Filters applied for Service, Urgency, Assignment Group, and SLA status                                                          |
| 4     | DAX Queries Used            | SLA Compliance % =<br>DIVIDE(COUNTROWS(FILTER(Incidents,Incidents[SLA_Met]=TRUE()))),COUNTROWS(Incidents))                      |
| 5     | Dashboard design            | Incident volume, SLA compliance, and escalation-tracking views.<br>No. of Visualizations / Graphs – 6                           |
| 6     | Report Design               | Summary reporting on incident resolution and knowledge-article reuse. No. of Visualizations / Graphs – 4                        |

5. Tableau — Performance Test

|               |                                             |
|---------------|---------------------------------------------|
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Model Performance Testing

| S.No. | Parameter               | Screenshot / Values                                                                                   |
|-------|-------------------------|-------------------------------------------------------------------------------------------------------|
| 1     | Data Rendered           | Incident, change, and child-incident records exported from ServiceNow via scheduled Table API extract |
| 2     | Data Preprocessing      | Grouping incidents by service, technician, and resolution status                                      |
| 3     | Utilization of Filters  | Filters for Service, Level 2 Assignment, and Resolution status                                        |
| 4     | Calculation fields Used | Days Open = DATEDIFF([Opened Date],[Resolved Date],'day')                                             |
| 5     | Dashboard design        | Incident-lifecycle tracking dashboard. No. of Visualizations / Graphs – 5                             |
| 6     | Story Design            | Story walking through incident creation → escalation → resolution. No. of Visualizations / Graphs – 4 |

## 6. GenAI Functional & Performance Testing

|               |                                             |
|---------------|---------------------------------------------|
| Date          | 08 September 2026                           |
| Team ID       |                                             |
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| Maximum Marks | 10 Marks                                    |

### Test Scenarios & Results

| Test Case ID | Scenario                                          | Steps                                                  | Expected Result                                                | Actual Result                                                       | Pass/Fail |
|--------------|---------------------------------------------------|--------------------------------------------------------|----------------------------------------------------------------|---------------------------------------------------------------------|-----------|
| FT-01        | Validate incident classification fields           | Enter valid/invalid Category, Subcategory, Urgency, CI | Valid inputs accepted; errors shown for invalid ones           | Invalid entries correctly blocked with inline error messages        | Pass      |
| FT-02        | Knowledge / Agent Assist suggestions              | Search a known issue                                   | Relevant knowledge articles returned                           | Top 3 matching articles displayed within the incident form          | Pass      |
| FT-03        | GenAI incident summary generation (Now Assist)    | Trigger "Generate Summary" on a resolved incident      | Concise, accurate summary of incident and resolution generated | Summary generated in ~4s, accurately reflected cause and resolution | Pass      |
| FT-04        | Escalation / reassignment                         | Reassign incident to Level 2                           | Incident routed to correct group                               | Incident appeared in Level 2 queue immediately after reassignment   | Pass      |
| PT-01        | Incident save time                                | Time saving a new incident                             | Completes within acceptable response time                      | Average save time 1.4s across 10 runs                               | Pass      |
| PT-02        | Concurrent incident load                          | Create several incidents at once                       | System does not slow down or fail                              | 20 simultaneous incident creations completed with no errors         | Pass      |
| PT-03        | Related-record load (attachments/child incidents) | Attach files / create child incidents                  | Processes smoothly, no crashes                                 | 5 attachments and 3 child incidents processed without issue         | Pass      |

7. User Acceptance Testing (UAT) Template

|               |                                             |
|---------------|---------------------------------------------|
| Date          | 08 September 2026                           |
| Team ID       |                                             |
| Project Name  | Incident Lifecycle Automation in ServiceNow |
| Maximum Marks | 4 Marks                                     |

Project Overview

UAT was performed on the Incident Lifecycle Automation solution in ServiceNow, covering service setup, incident creation/classification, knowledge-assisted investigation, Level 2 escalation, emergency changes, child incidents, resolution, and knowledge-article creation.

Project Version: v1.0 | Testing Period: 02 Sept 2026 to 08 Sept 2026

Testing Scope

Service setup · Incident creation & classification · Knowledge/Agent Assist · Escalation to Level 2 · Emergency Change · Child Incident · Resolution & documentation · Knowledge Article creation · SLA tracking · Final validation

Test Cases

| ID     | Scenario                   | Steps                         | Expected                | Actual                   | P/F  |
|--------|----------------------------|-------------------------------|-------------------------|--------------------------|------|
| TC-001 | Create & classify incident | Set Category/Urgency/CI; save | Correctly classified    | Saved correctly          | Pass |
| TC-002 | Knowledge/Agent Assist     | Search knowledge base         | Relevant articles shown | 3 articles surfaced      | Pass |
| TC-003 | Escalate to L2             | Reassign to L2 group          | Routed to L2            | Visible in L2 queue      | Pass |
| TC-004 | Emergency Change           | Create linked Change          | Linked correctly        | Change created & linked  | Pass |
| TC-005 | Child Incident             | Create linked child           | Linked correctly        | Linked; context retained | Pass |
| TC-006 | Resolve incident           | Enter cause; close            | Resolved & documented   | Captured; closed         | Pass |
| TC-007 | Knowledge Article          | Convert incident              | Article created         | Published & linked       | Pass |

Bug Tracking

| Bug ID | Description                                                       | Severity | Status |
|--------|-------------------------------------------------------------------|----------|--------|
| BG-001 | Knowledge Article link showed only after page refresh             | Low      | Closed |
| BG-002 | Emergency Change field was editable after the incident was closed | Medium   | Closed |

**Notes:** All test cases passed; two low/medium-severity issues found and resolved. Recommended for deployment.